

SQL Analysis for Online Book Store

Extracting business insights from sales data.



Project Objectives

Customer Behavior

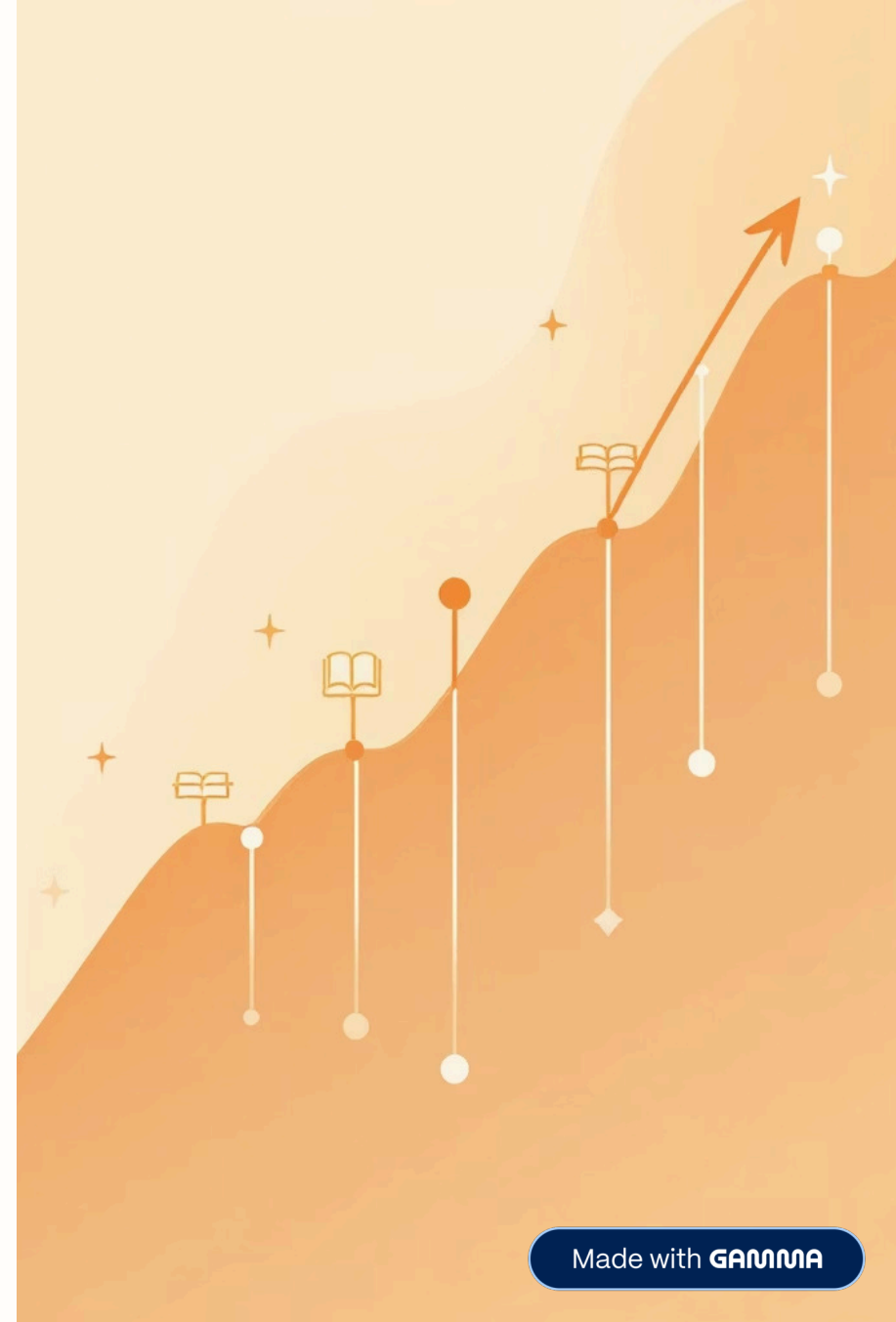
Understand purchasing patterns.

Best-Selling Books

Identify top performers.

Revenue & Trends

Track financial performance and category sales.



SQL in Action: Data Exploration

Key Reports Generated

- Total Sales
- Top Customers
- Popular Categories
- Monthly Revenue
- Order Frequency





Improving Business Decisions



Better Decisions



Efficient Inventory



Targeted Marketing

Core SQL Queries: Books & Customers

Fiction Books

Retrieve all books in the "Fiction" genre, ordered by price.

```
SELECT * FROM book WHERE genre='Fiction' ORDER BY price DESC LIMIT 5;
```

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.89	29
2	260	Business-focused methodical applicati...	Brian King	Fiction	1907	49.59	10
3	142	Multi-tiered responsive parallelism	Amanda Wilson	Fiction	1940	48.96	11
4	274	Automated systemic functionalities	Matthew Thomas	Fiction	1903	48.80	72
5	278	Exclusive asymmetric installation	Benjamin Flores	Fiction	1901	48.61	69

Post-1950 Books

Find books published after the year 1950.

```
SELECT * FROM book WHERE  
published_year > 1950 LIMIT 5;
```

	book_id [PK] integer	title character varying	author character varying	genre character varying	published_year integer	price numeric (10,2)	stock integer
1	2	Persevering reciprocal knowledge u...	Mario Moore	Fantasy	1971	35.80	19
2	4	Customizable 24hour product	Christopher Andre...	Fiction	2020	43.52	8
3	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
4	6	Advanced encompassing implemen...	Bryan Morgan	Biography	1985	6.56	2
5	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84

Customer & Order Insights

Canadian Customers

List all customers from Canada.

```
SELECT * FROM customer WHERE country='Canada';
```

	customer_id [PK] integer	name character varying	email text	phone numeric	city character varying	country character varying
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.c...	1234568358	Thompsonfurt	Canada

November 2023 Orders

Show orders placed in November 2023,
by total amount.

```
SELECT * FROM orders WHERE  
order_date BETWEEN '2023-11-01'  
AND '2023-11-30' ORDER BY  
total_amount DESC LIMIT 5;
```

	order_id integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	245	386	97	2023-11-01	9	411.66
2	137	474	471	2023-11-25	8	363.04
3	19	496	60	2023-11-17	9	316.26
4	4	433	343	2023-11-25	7	301.21
5	213	325	447	2023-11-17	7	253.75

Inventory & Pricing

Total Book Stock

Retrieve the total stock of books available.

```
SELECT SUM(stock) FROM book;
```

	sum bigint 
1	25056

Most Expensive Book

Find details of the most expensive book.

```
SELECT * FROM book ORDER BY  
price DESC LIMIT 1;
```

	book_id [PK] integer 	title character varying 	author character varying 	genre character varying 	published_year integer 	price numeric (10,2) 	stock integer 
1	340	Proactive system-worthy orchestrati...	Robert Scott	Mystery	1907	49.98	88

Order Value & Genres

Orders Over \$20

Retrieve all orders where the total amount exceeds \$20.

```
SELECT * FROM orders WHERE total_amount > 20;
```

	order_id integer 🔒	customer_id integer 🔒	book_id integer 🔒	order_date date 🔒	quantity integer 🔒	total_amount numeric (10,2) 🔒
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

Available Genres

List all distinct genres available in the Books table.

```
SELECT DISTINCT genre FROM book;
```

	genre character varying 🔒
1	Romance
2	Biography
3	Mystery
4	Fantasy
5	Fiction

Advanced Insights: Revenue & Sales

Total Revenue

Calculate total revenue generated from all orders.

```
SELECT SUM(total_amount) AS Revenue FROM orders;
```

	revenue numeric
1	75628.66

Books Sold by Genre

Retrieve total number of books sold for each genre.

```
SELECT book.genre,  
SUM(orders.quantity) AS Total_Sold  
FROM book JOIN orders ON  
orders.book_id=book.book_id  
GROUP BY 1 ORDER BY Total_Sold;
```

	genre character varying	total_sold bigint
1	Fiction	225
2	Biography	285
3	Non-Fiction	351
4	Romance	439
5	Fantasy	446

Customer Spending & Stock Management

Top Spender

Find the customer who spent the most on orders.

```
SELECT customer.customer_id, customer.name, SUM(orders.total_amount)
Amount_Spend FROM customer JOIN orders ON
customer.customer_id=orders.customer_id GROUP BY 1,2 ORDER BY
Amount_Spend DESC LIMIT 1;
```

	customer_id [PK] integer	name character varying	amount_spend numeric
1	457	Kim Turner	1398.90

Remaining Stock

Calculate stock remaining after fulfilling all orders.

```
SELECT book.book_id, book.title,
book.stock,
COALESCE(SUM(orders.quantity),0)
AS Order_Quantity, book.stock-
COALESCE(SUM(orders.quantity),0)
AS Remaining_Stock FROM book
LEFT JOIN orders ON
book.book_id=orders.book_id
GROUP BY 1 ORDER BY
book.book_id;
```

	book_id [PK] integer	title character varying	stock integer	order_quantity bigint	remaining_stock bigint
1	1	Configurable modular throughput	100	3	97
2	2	Persevering reciprocal knowledg...	19	0	19
3	3	Streamlined coherent initiative	27	5	22
4	4	Customizable 24hour product	8	0	8
5	5	Adaptive 5thgeneration encoding	16	8	8